

Geometry of Root Systems and Dynkin Diagrams

Classification of Continuous Symmetries in Physics

Theoretical Physics Master - Group Theory

Master Study Guide

The Roadmap: From Algebra to Geometry

Main Goal: To demonstrate how the classification of semi-simple Lie algebras reduces to the geometric tessellation of a Euclidean space.

Logical Path (30 Minutes):

1. **The Foundation:** Assume the Cartan-Weyl decomposition and the Euclidean nature of the root space.
2. **The Geometric Constraint:** Derive the integer quantization of angles (Crystallographic condition).
3. **Rank-2 Classification:** Visually construct the only 4 allowed 2D root systems ($A_1 \oplus A_1, A_2, B_2, G_2$).
4. **Dynkin Diagrams & The End Result:** Define simple roots, extract Dynkin diagrams, and state the final classification (ADE and Exceptionals), including accidental isomorphisms.

The Starting Point: Roots in Euclidean Space

Assumed Structure: Let $\mathfrak{g}$ be a complex semi-simple Lie algebra. We assume the Cartan-Weyl decomposition:

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \mathcal{R}} \mathfrak{g}_{\alpha}$$

The roots $\alpha \in \mathcal{R}$ live in a real Euclidean space $E = \mathfrak{h}_{\mathbb{R}}^{\vee}$, equipped with a scalar product $(\cdot, \cdot)$ inherited from the Killing form.

Physical Meaning: The roots represent the gauge bosons (like $W^{\pm}$, gluons) that mediate transitions between quantum states charged under the Cartan generators.

Formal Definition 4.4.1: Root System

The Crystallographic Restriction

The representation theory of the $\mathfrak{su}(2)$ subalgebras associated with each root imposes a severe geometric constraint: the projection of any root onto another must be a half-integer.

$$\forall \alpha, \beta \in \mathcal{R} : \quad n_{\alpha\beta} \equiv \frac{2(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z} \quad (1)$$

By expressing this in terms of the angle $\theta_{\alpha\beta}$ and norms:

$$\begin{aligned} n_{\alpha\beta} n_{\beta\alpha} &= \left(\frac{2(\alpha, \beta)}{(\alpha, \alpha)} \right) \left(\frac{2(\beta, \alpha)}{(\beta, \beta)} \right) \\ &= 4 \frac{(\alpha, \beta)^2}{\|\alpha\|^2 \|\beta\|^2} = 4 \cos^2 \theta_{\alpha\beta} \in \mathbb{Z} \end{aligned} \quad (2)$$

Extended Proof of Angle Quantization

Allowed Angles and Length Ratios

Since $0 \leq \cos^2 \theta \leq 1$, the integer constraint (2) restricts us to $n_{\alpha\beta}n_{\beta\alpha} \in \{0, 1, 2, 3, 4\}$.

Excluding 4 (which implies collinear roots $\beta = \pm\alpha$, trivial by Def. 4.4.1), the allowed obtuse angles ($\theta \geq \pi/2$) for linearly independent roots are:

- ▶ **Orthogonal:** $\theta = \pi/2 \implies \|\beta\|/\|\alpha\|$ is undetermined.
- ▶ **Equal length:** $\theta = 2\pi/3$ (120°) $\implies \|\beta\|/\|\alpha\| = 1$.
- ▶ **Ratio $\sqrt{2}$:** $\theta = 3\pi/4$ (135°) $\implies \|\beta\|/\|\alpha\| = \sqrt{2}$.
- ▶ **Ratio $\sqrt{3}$:** $\theta = 5\pi/6$ (150°) $\implies \|\beta\|/\|\alpha\| = \sqrt{3}$.

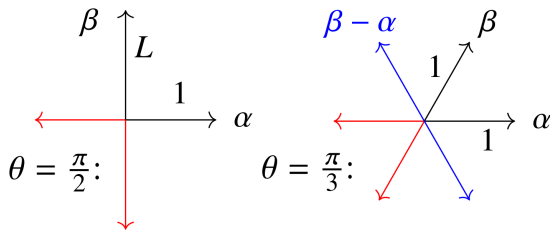
$n_{\alpha\beta}n_{\beta\alpha}$	3	2	1	0	1	2	3
$n_{\beta\alpha}$	1	1	1	0	-1	-1	-1
$n_{\alpha\beta}$	3	2	1	?	-1	-2	-3
$\ \beta\ /\ \alpha\ $	$\sqrt{3}$	$\sqrt{2}$	1	?	1	$\sqrt{2}$	$\sqrt{3}$

Constructing Rank-2 Root Systems

In a 2-dimensional root space ($rank = 2$), we pick two roots α, β spanning E and saturate the space by applying:

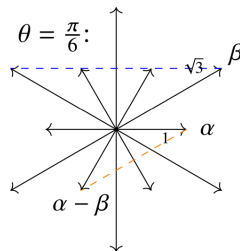
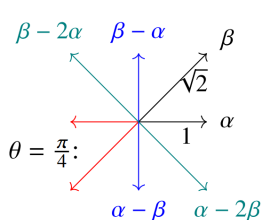
1. Weyl reflections $s_\alpha(\beta) = \beta - n_{\alpha\beta}\alpha$.
2. The α -string property (Thm 4.4.2): if $(\alpha, \beta) > 0$, then $\alpha - \beta \in \mathcal{R}$.

This purely geometric construction yields exactly 4 systems:



Rank-2 Root Systems: B_2 and G_2

For the remaining allowed angles, the reflection symmetries generate highly structured lattices:



Theorem 4.4.2: Root Strings

Simple Roots & Dynkin Diagrams

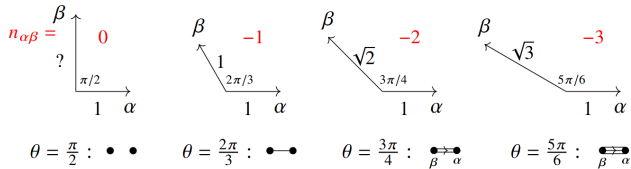
To generalize beyond $rank = 2$, we must identify the "basis" vectors. We define **Simple Roots** ($\Delta \subset \mathcal{R}$) as positive roots that cannot be decomposed into a sum of other positive roots.

Definition 4.4.5 (Dynkin Diagrams): A graph encoding the geometry of Δ :

- ▶ **Nodes:** Each simple root is a node.
- ▶ **Edges:** Nodes are connected by $\ell_{\alpha\beta} = 4 \cos^2 \theta_{\alpha\beta} \in \{0, 1, 2, 3\}$ lines.
- ▶ **Arrows:** If lengths differ ($\ell > 1$), an arrow points from the longer root to the shorter one ($>$).

Formal Def 4.4.3: Simple Roots

Dynkin Diagrams for Rank-2



These 4 fundamental building blocks encode the entire Cartan matrix $C_{ij} = n_{\alpha(j)}\alpha(i)$.

The Final Classification (A,B,C,D,E,F,G)

By classifying all "admissible" diagrams (excluding loops and vertices with > 3 lines to avoid violating Cauchy-Schwarz), we obtain the complete list of simple Lie algebras:

- ▶ **Classical Series:** $A_n(\mathfrak{su}_{n+1})$, $B_n(\mathfrak{so}_{2n+1})$, $C_n(\mathfrak{sp}_{2n})$, $D_n(\mathfrak{so}_{2n})$.
- ▶ **Exceptional Algebras:** G_2 , F_4 , E_6 , E_7 , E_8 .

Dynkin diagram	$\mathcal{R}$ -sys.	$\mathfrak{g}_{\mathbb{C}}$	Cartan matrix C_{ij}
	A_n	$\mathfrak{su}(n+1)_{\mathbb{C}}$	see Def. 3.6.11
	B_n	$\mathfrak{so}(2n+1)_{\mathbb{C}}$	see Def. 3.6.11
	C_n	$\mathfrak{sp}(2n)_{\mathbb{C}}$	see Def. 3.6.11
	D_n	$\mathfrak{so}(2n)_{\mathbb{C}}$	see Def. 3.6.11
	G_2	$(\mathfrak{g}_2)_{\mathbb{C}}$	$\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$
	F_4	$(\mathfrak{f}_4)_{\mathbb{C}}$	$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & 2 & 2 & -1 \end{pmatrix}$

Accidental Isomorphisms

Dynkin diagrams provide an immediate visual proof of "accidental isomorphisms" between seemingly different matrix groups at low rank:

- ▶ $A_1 \cong B_1 \cong C_1 \iff \mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{so}(3)_{\mathbb{C}} \cong \mathfrak{sp}(2)_{\mathbb{C}}$
- ▶ $B_2 \cong C_2 \iff \mathfrak{so}(5)_{\mathbb{C}} \cong \mathfrak{sp}(4)_{\mathbb{C}}$
- ▶ $A_1 \oplus A_1 \cong D_2 \iff \mathfrak{su}(2)_{\mathbb{C}} \oplus \mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{so}(4)_{\mathbb{C}}$
- ▶ $A_3 \cong D_3 \iff \mathfrak{su}(4)_{\mathbb{C}} \cong \mathfrak{so}(6)_{\mathbb{C}}$

$$\bullet \rightsquigarrow A_1 \cong B_1 \cong C_1 \iff \mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{so}(3)_{\mathbb{C}} \cong \mathfrak{sp}(2)_{\mathbb{C}}.$$

$$B_2 \cong \bullet \rightleftarrows \bullet \cong \bullet \rightleftarrows \bullet \cong C_2 \iff \mathfrak{so}(5)_{\mathbb{C}} \cong \mathfrak{sp}(4)_{\mathbb{C}}.$$

$$A_1 \oplus A_1 \cong \bullet \cdot \bullet \cong D_2 \iff \mathfrak{su}(2)_{\mathbb{C}} \oplus \mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{so}(4)_{\mathbb{C}}.$$

$$D_3 \cong \bullet \cdot \bullet \cdot \bullet \cong \bullet \text{---} \bullet \text{---} \bullet \cong A_3 \iff \mathfrak{so}(6)_{\mathbb{C}} \cong \mathfrak{su}(4)_{\mathbb{C}}.$$

Definition 4.4.1: Root System

Let E be a finite-dimensional Euclidean vector space, with $(\cdot, \cdot) : E \times E \rightarrow \mathbb{R}$ the scalar product. A root system $\mathcal{R} \subset E$ is a finite set which satisfies:

- a) The $\mathbb{R}$ -span of $\mathcal{R}$ is E .
- b) For $\alpha \in \mathcal{R}$, if $\lambda\alpha \in \mathcal{R}$ for $\lambda \in \mathbb{R}$, then $\lambda = \pm 1$.
- c) **Crystallographic condition:**

$$\forall \alpha, \beta \in \mathcal{R} : \quad n_{\alpha\beta} := \frac{2(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z}.$$
- d) **Reflection invariance:**

$$\forall \alpha, \beta \in \mathcal{R} : \quad s_{\alpha}(\beta) := \beta - n_{\alpha\beta}\alpha \in \mathcal{R}.$$

Extended Proof: Angle Quantization

From Definition 4.4.1 (c), we know $n_{\alpha\beta} \in \mathbb{Z}$. Using Euclidean geometry definitions:

$$\begin{aligned}\|\alpha\| &= \sqrt{(\alpha, \alpha)} \\ \cos \theta_{\alpha\beta} &= \frac{(\alpha, \beta)}{\|\alpha\| \|\beta\|}\end{aligned}$$

Substitute (α, β) into the product of integers:

$$\begin{aligned}n_{\alpha\beta} n_{\beta\alpha} &= \left(2 \cos \theta_{\alpha\beta} \frac{\|\beta\|}{\|\alpha\|} \right) \left(2 \cos \theta_{\alpha\beta} \frac{\|\alpha\|}{\|\beta\|} \right) \\ &= 4 \cos^2 \theta_{\alpha\beta}\end{aligned}$$

Since $\cos^2 \theta \in [0, 1]$, the integer product must be $\in \{0, 1, 2, 3, 4\}$.

Theorem 4.4.2: Root Strings

For $\alpha, \beta \neq \alpha$ two roots, the α -string through β (the set $\beta - p\alpha, \dots, \beta + q\alpha$) satisfies:

- a) The α -string through α is $\{-\alpha, 0, \alpha\}$.
- b) $p - q = n_{\alpha\beta} \equiv \frac{2(\alpha, \beta)}{(\alpha, \alpha)}$.
- c) For $\alpha \neq \pm\beta$, the string has at most 4 elements ($p + q \leq 3$).
- d) **Crucial property for drawings:** If $(\alpha, \beta) > 0$, then $\alpha - \beta \in \mathcal{R}$. If $(\alpha, \beta) < 0$, then $\alpha + \beta \in \mathcal{R}$.

Definition 4.4.3: Positive & Simple Roots

Let $t \in E$ be such that $(t, \alpha) \neq 0$ for all $\alpha \in \mathcal{R}$ (always possible because $\mathcal{R}$ is finite).

- ▶ A root $\alpha \in \mathcal{R}$ is called **positive** ($\alpha \in \mathcal{R}^+$) if $(t, \alpha) > 0$, and **negative** if $(t, \alpha) < 0$.
- ▶ A positive root is called **simple** if it cannot be written as the sum of two positive roots.

The set of simple roots is denoted $\Delta \subset \mathcal{R}$.

Lemma 4.4.4: If $\alpha, \beta \in \Delta$, then: a) $\alpha - \beta$ and $\beta - \alpha$ are not roots. b) $(\alpha, \beta) \leq 0$ (obtuse angles only). c) Δ is linearly independent.